

ASSIGNMENT ON CSE 420

Submitted to: Sajid (SZD) Sir

Submitted by: Sheikh Md. Tahmid Hossain

ID : 22299522

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Section: 12

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Assignment No.2

Grammar

$E \rightarrow FF$

$F \rightarrow pF$

$F \rightarrow q$

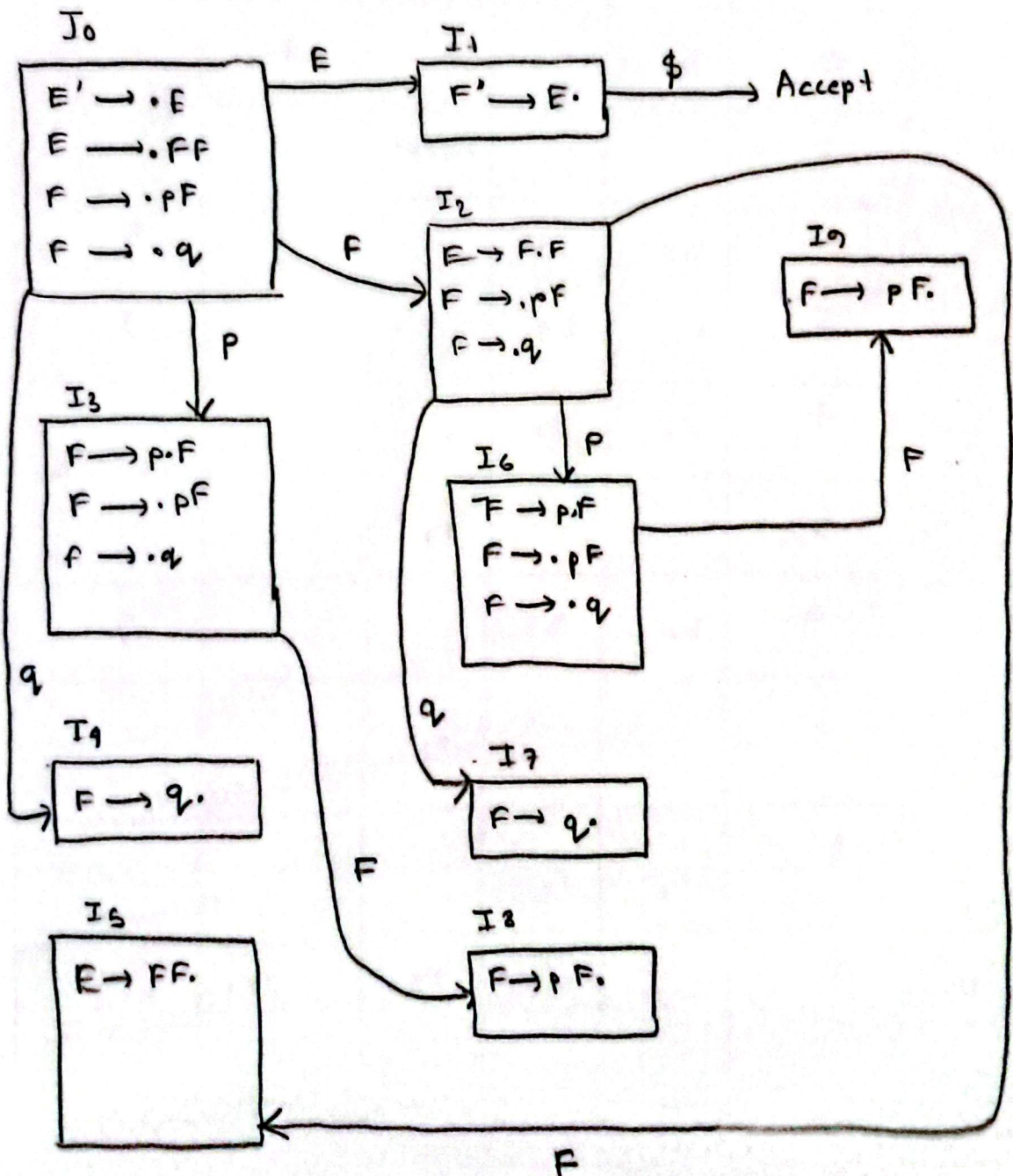
→
Augmentation

$E' \rightarrow E$

$E \rightarrow FF$

$F \rightarrow pF$

$F \rightarrow q$



First [F and follow's

$$F = \{p, q\}$$

$$E = \{p, q\}$$

Parsing Table

state	p	q	\$	E	F
0	s ₃	s ₄		1	2
1			Accept		
2	s ₆	s ₇			5
3	s ₃	s ₄			8
4	r ₃	r ₃			
5			r ₁		
6	s ₆	s ₇			9
7			r ₃		
8	r ₂	r ₁			
9			r ₂		

Parsing String 'PPPP q'

Step	State	Input	Action
1	0	PPP PPq \$	S3
2	03	PPPP q\$	S3
3	033	PPPq \$	S3
4	0333	PPq \$	S3
5	03333	Pq \$	S3
6	033333	q \$	S4
7	0333334	\$	$r_3: F \rightarrow q$
8	0333338	\$	$r_2: F \rightarrow pF$
9	03338	\$	$r_2: F \rightarrow pF$
10	0338	\$	$r_2: F \rightarrow pF$
11	0338	\$	$r_2: F \rightarrow pF$
12	038	\$	$r_2: F \rightarrow pF$
13	08	\$	Action [2, \$) is empty which results an error

so, the string is rejected, not
accepted.